

When Does Soccer Need Mixed Strategies?

A pure-first Nash-Q approach: where
a two-player soccer Markov game
does — and does not — need mixing,
and how much LP work that saves.

FOUR QUESTIONS

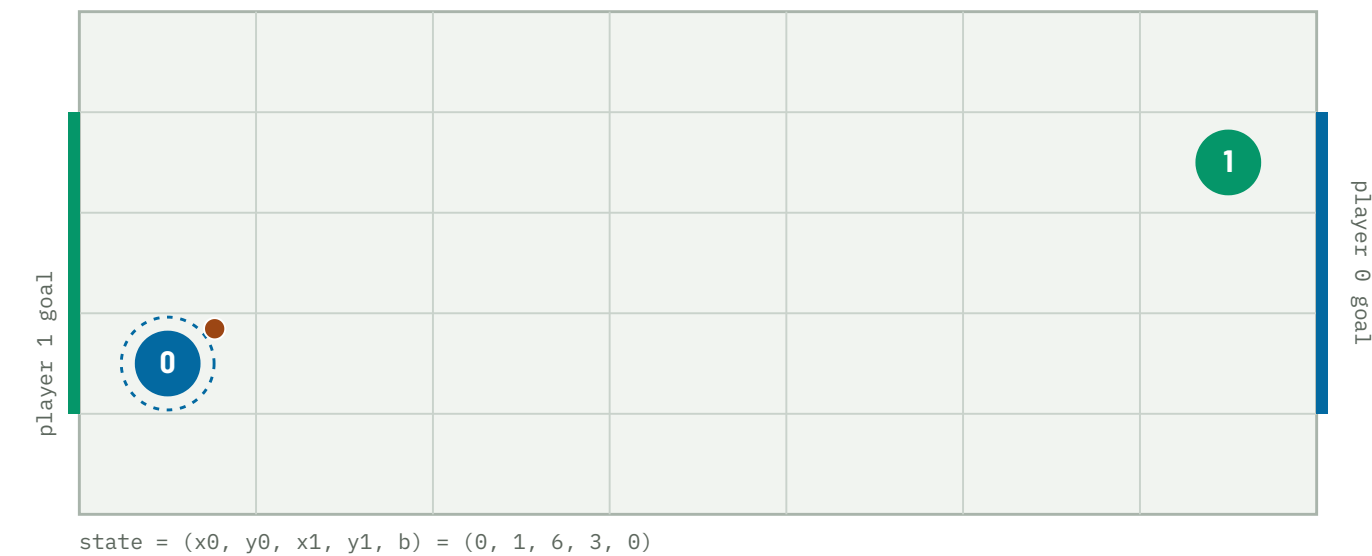
RQ1 — when does the soccer Markov game admit pure versus mixed stage equilibria?

RQ2 — can pure-saddle detection reduce Nash-Q computation without changing the exact value?

RQ3 — what spatial and transition structures characterize the mixed-equilibrium states?

RQ4 — how sensitive are classification and computation to discounting, tolerance, and reward formulation?

repo · soccer-markov-nash | 315+ tests | game family in docs/design.md | methods in docs/methods.md | every policy drawn in docs/gallery.html



The kickoff. 7×5 grid, 2 380 non-terminal states. Player 0 (blue) carries the ball and attacks the right edge; player 1 (green) the left. Start positions are the ones the

A10 page assigns this netID. Actions U D L R are chosen simultaneously; reward is +1 / -1 / 0, zero-sum.

POSITION RELATIVE TO LITTMAN (1994)

Littman **already established** that the soccer Markov game can require mixed strategies — his Figure 2 gives a state where any deterministic offensive choice is blocked indefinitely and only randomising between stand and move guarantees an opening. This project does not rediscover that. It asks: **where does mixing appear, why does it appear, and what parameter switches it on?** Littman demonstrated the phenomenon in a *particular* soccer Markov game; we investigate which structural features determine when mixed equilibria arise, and whether their computational cost can be avoided except where mixing is genuinely necessary (docs/littman.md reproduces his Figure 2 and Table 3).

ABSTRACT

- 1** The soccer game returns pure strategies until the collision rule couples the ball-steal to both players' actions. Deterministic or coin-flip resolution → a pure saddle at every one of the 238 000 (state×step) stage games (also positionally determined). Littman's random *move order* → mixing, but only when it is the majority rule (sharp threshold at $\text{blend} \approx 0.5$ in this interpolation family) and only with a multi-cell goal. That last clause is specific to the move-order rule: action-independent movement noise (slip) forces mixing at any goal width.
- 2** The pure-first hybrid eliminates LP work on almost every stage game — 100% of states on the deterministic game, 96.05% on the random one — matching the all-LP value function to $4e-16$. That is a property of the game; the wall-clock speedup is reported separately.
- 3** 94 no-pure-saddle states are not 94 different situations — they are 8 geometric templates, all within 2 cells of the opponent. The mixed states combine stochastic move order, a multi-cell goal mouth, and a defender with insufficient one-step coverage; *every* tested mixed stage game contains a 2×2 matching-pennies submatrix. Canonicalized under the board mirror they collapse to 47 pairs, then by carrier-frame geometry to 8 templates; every one occurs at Manhattan distance 1 or 2 between the players (40 at distance 1, 54 at distance 2, zero at distance ≥ 3). **Mixing is spatially localized around the opponent: all 94 no-pure-saddle states occur at player distance 1 or 2, while goal-mouth geometry determines whether competing scoring lanes exist in the first place.** The single-cell → pure / wider → mixed switch is machine-certified invariant to the discount, the action set, and the reward objective (make verify); the region carries 41% of the equilibrium-path occupancy at 4% of the state space.

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- 4** **Action accuracy is not equilibrium accuracy.** A policy network that names the exact-optimal action 99.9% of the time is more exploitable than a mediocre value network. On the harder game where mixed equilibria genuinely matter, it gets sharper still: fitting a Q-net directly to the exact stage matrices is the closest numerical fit and the single most exploitable policy of the four baselines tested — matching a continuous target well and matching a discrete equilibrium well are different objectives. The exact solver stays the ground truth.
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READING GUIDE: CORE, SUPPORTING, EXPLORATORY

This report covers more ground than its central claim needs (full accounting in §11, “What weight each experiment carries”). The core story is one chain:

where mixing occurs → why it occurs → the pure-first exact solver → why mixing matters against a best response

If time is short, follow exactly that, in that order:

- **Core — Where:** the 94 no-pure-saddle states, all at Manhattan distance 1–2 between the players, clustering into 8 geometric templates (RQ3). **Why:** a stochastic move order plus a multi-cell goal produces crossing best replies — a certified 2×2 matching-pennies core at every one (Mechanism). **The solver:** the pure-first hybrid backup, exact everywhere, eliminating LP work on 96–100% of stage games (Method, RQ2). **Why it matters:** a minimax policy exploits weak opponents and survives a challenger built specifically to beat it, while every deterministic policy — including a hand-scripted one that beats a random opponent ~76% of the time — does one or the other, never both (“Why it matters” below). That last result is the direct answer to *why any of this matters*: a predictable pure policy can be learned and exploited, and the 94 no-pure-saddle states are exactly where that predictability would show up.
- **Supporting validation** — numerical robustness across discount and tolerance (RQ4); self-play and occupancy on the equilibrium path (docs/occupancy.md); mirror symmetry (value *and* policy equivariance); which no-pure-saddle states are a uniquely forced mix vs. degenerate — the 64/16/14 split (docs/degeneracy.md).

- **Exploratory extensions**, groundwork for future work, not equal-weight contributions — neural function approximation, including on the harder random-move-order game; general-sum solving; reward shaping; policy iteration; the single-cell proof machinery; the phase diagram; the transition-rule variants (tackle, slip, blend). The eventual actual challenge — a continuous-action version of this game — is future work in the same sense; see the roadmap at the end of the Exploratory section.

Method

THREE CONTRIBUTIONS, AND THE HYPOTHESIS

C1 — Computational: a pure-first hybrid Nash-Q backup — check for a pure saddle ($O(A^2)$), use it, LP only where it fails; exact everywhere; eliminates LP work on 96–100% of stage games. **C2 — Structural:** characterization of where the no-pure-saddle states occur (over the tested parameter grid, not a theorem) — the goal-width switch, local player geometry, the 8 geometric templates, and the certified matching-pennies mechanism. **C3 — Practical:** the strategically mixed region matters disproportionately under equilibrium play and specifically prevents exploitation by a best-response opponent — not merely a validation check (§7).

Hypothesis: mixed equilibria are not distributed arbitrarily — they arise from a specific interaction between goal-mouth geometry and stochastic move-order coupling, and because most states keep pure saddles a pure-first solver gets the exact value while substantially cutting LP usage.

OBJECTIVE DEFINITIONS – RELATED BUT NOT INTERCHANGEABLE

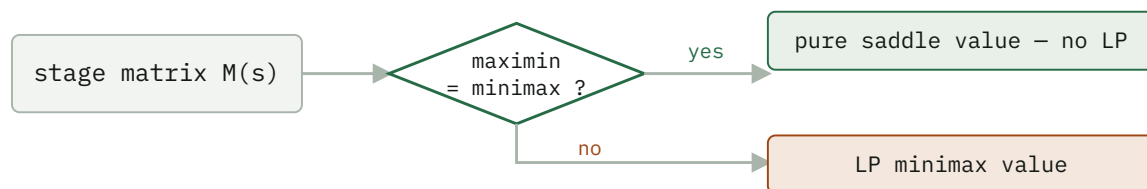
WHERE	OBJECTIVE
A10 Part 1 / exact	undiscounted, finite 100-step horizon (run_finite_horizon, non-stationary v)
stationary Nash-Q	discounted infinite-horizon fixed point, $\gamma < 1$ (run()) – a contraction proxy for the above
scoring="rate"	continuing discounted: a goal resets, value = expected discounted goal difference
Littman replication	Littman's discount / draw interpretation, $\gamma = 0.9 \equiv 0.1$ per-step draw probability

Unless a section says otherwise, results are the stationary discounted fixed point at $\gamma = 0.9$; RQ4 shows the pure/mixed classification is stable across $\gamma \in [0.5, 0.995]$ and matches the exact undiscounted answer.

The game is zero-sum, so at every state the stage game is a payoff matrix $M(s) [a_0, a_1] = E[R + \gamma V(s')]$ and the Shapley (1953) fixed point is the minimax value:

$$\begin{aligned} V(s) &= \text{val}(M(s)) = \max_n \min_{k \in \mathcal{K}} (p^T M) [a_1] \\ &= \min_p \max_{a_0} (Mq) [a_0] \end{aligned}$$

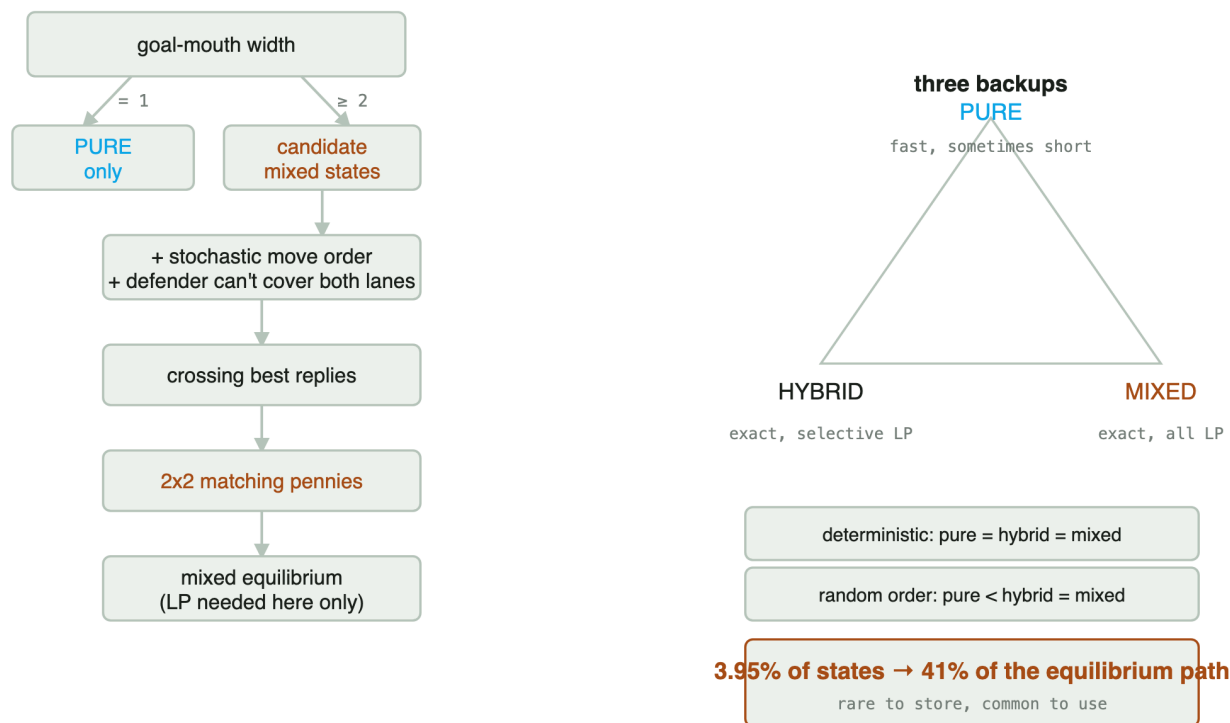
The $Q = R + \beta \text{Nash}(Q')$ update is this operator. The A10 game is *undiscounted* (± 1 at a goal, tie after 100 steps); $\gamma < 1$ is a contraction device for value iteration and RQ4 shows the split holds across γ . Three stage backups: **pure** (the maximin security value — a fast lower bound, not a Nash value when it is below the minimax), **mixed** (the LP minimax value, via scipy HiGHS), and **hybrid** (the pure saddle value where maximin = minimax, the LP elsewhere — exact everywhere). Support enumeration (support_enum.py) finds *all* equilibria, including of general-sum games.



The pure-first hybrid. On the deterministic game the "yes" branch is taken at every one of the 2380 states, so the LP is never called.

RESULTS

Where mixing comes from, and what the solver does about it



The whole story. Left: the mechanism chain – goal width, then stochastic move order and interception geometry, produce crossing best replies and a matching-pennies stage game where the LP is needed. Right: the three backups, the experimental result, and the occupancy headline.

RQ1 – pure versus mixed, by collision rule

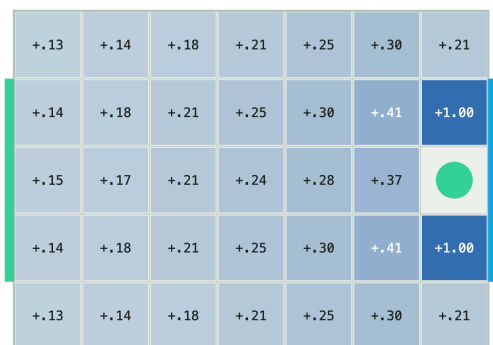
The A10 game is undiscounted, so it is solved exactly by 100-step backward induction (run_finite_horizon, experiments/undiscounted.csv):

EXACT UNDISCOUNTED GAME – 238 000 (STATE × STEP) STAGE GAMES

MOVE ORDER	MIXED STAGE GAMES	V(KICKOFF)	PURE EQUILIBRIUM?
deterministic (exact A10)	0	0.000	yes – pure (non-stationary)
coinflip tie-break	0	0.000	yes
random move order	3 148 (404 states)	+0.459	no

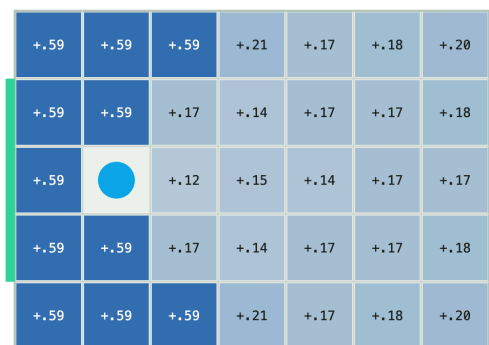
The deterministic equilibrium is **constructive and memoryless**: each player's win attractor (states from which it forces a goal, soccer_nash/attractor.py) equals its $V^* = \pm 1$ set, and the pure profile "attractor move on your winning set, safety move elsewhere" realizes V^* at every state (scripts/positional.py) — the game is positionally determined.

carrier position (player 0)



V^* by player 0 position (blue = player 0 ahead)

defender position (player 1)



V^* by player 1 position (blue = player 0 ahead)

The **value surface** of the random-order game ($\gamma = 0.9$). Left: V^* by carrier position — a smooth gradient toward the attacking goal, saturating to +1 on the two cells that score in one move. Right: V^* by defender position with the carrier held near midfield — nearly flat, the defender's location barely moves the value.

The stationary $\gamma < 1$ solve agrees and is faster — 0 no-pure-saddle stage games on the deterministic game at every γ from 0.5 to 0.995, and 94 / 2380 on the random game at $\gamma = 0.9$ ($V(\text{kickoff}) = +0.164$ at the netID start). The no-pure-saddle count never depends on the kickoff.

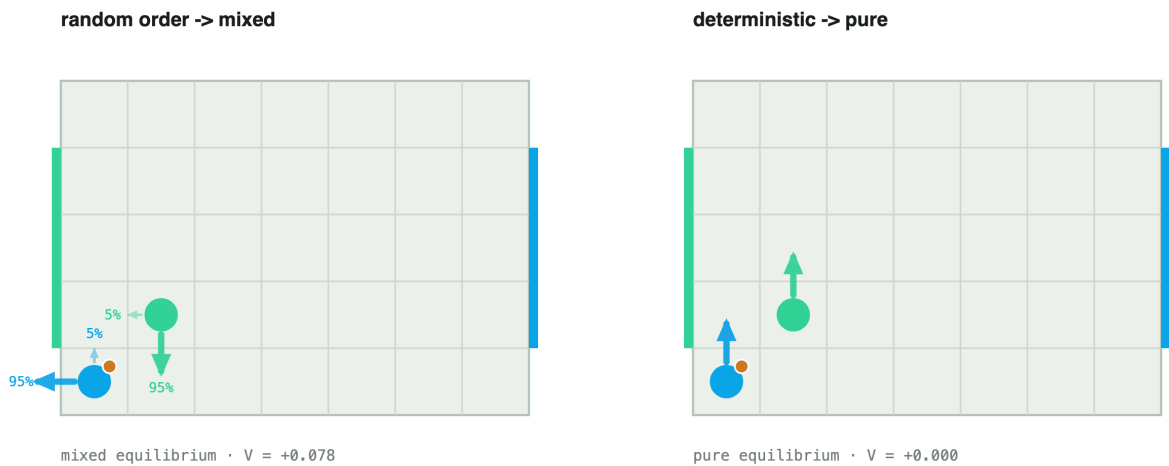
IT IS THE MOVE ORDER, NOT THE RANDOMNESS — AND IT HAS TO BE THE MAJORITY

The deterministic and coin-flip value functions are *identical* ($\max |V_{\text{det}} - V_{\text{coin}}| = 0$ at every γ): optimal play never enters a losing contest, so the coin is never flipped. Only Littman's random move order — where a ball-steal depends on who resolves first *and* on both players' targets — creates matching-pennies subgames. Interpolating the two (move_order="blend", resolve by random order with probability blend) shows the onset is **sharp: 0** mixed stage games for $\text{blend} \leq 0.5$, dozens at $\text{blend} = 0.52$, and an *overshoot* to $\sim 1.5\times$ the fully-random count near $\text{blend} \approx 0.7$ before it relaxes (scripts/blend.py, docs/blend.md).

Scope of this claim: in this particular soccer family, this one coin-flip collision variant does not create new mixed stage games, whereas random move ordering does — and only when it is the majority rule in the $P = (1-p)P_{\text{det}} + pP_{\text{random}}$ interpolation. This is *not* a statement about all action-independent stochastic transitions, which can certainly alter equilibrium structure in other games. Whether $p_c \approx 0.5$ is structural is open.



A threshold, not a ramp. No-pure-saddle stage games vs. the fraction of moves resolved by random order (5x4 board). Nothing until the random rule is the majority; then a jump and an overshoot.



One state, two resolution rules. Arrows are the equilibrium action distribution, sized by probability. Random order (Littman) makes the stage game matching pennies; deterministic carrier-wins collapses it to a pure best reply. Full set: [docs/gallery.html](https://docs.google.com/gallery).

RQ2 – Efficiency

Two numbers, kept apart: the **LP-call rate** is a property of the game and reproduces exactly; **wall clock** is a median of 5 repeats and depends on the machine and LP backend (scripts/benchmark.py).

RANDOM MOVE ORDER, $\Gamma = 0.9$ — THE CASE WHERE THE LP IS ACTUALLY NEEDED

SOLVER	SWEEPS	LP CALLS	STATES NEEDING LP	MEDIAN WALL CLOCK
pure (lower bound)	18	0	—	0.3 s
hybrid exact	108	9 672	3.95%	13.2 s
all-LP exact	108	~3.6 M	100%	~8 min

The portable claim: the hybrid eliminates LP work on almost every stage game — 100% of states on the deterministic game, **96.05%** on the random one — and matches the all-LP value to $4e-16$. That is a property of the game, not the machine. The observed wall-clock ratio (~13 s vs ~8 min) is reported separately: it also reflects matrix construction and the LP backend. **Mirror symmetry** (`run_symmetric`, deterministic dynamics only): every state has a distinct board-flip-plus-player-swap mirror, so the 2380 states are 1190 pairs — reconstruct one from the other by $V(\text{mirror}(s)) = -V(s)$ (and the equilibrium *policies* are equivariant too, where unique: `verify_policy_equivariance` returns 0 for the random game's 56 no-pure-saddle states).

3.95% OF STATES, BUT 41% OF THE EQUILIBRIUM PATH

The LP-call rate counts the whole state space. Under the Nash policy from the kickoff only **456 of 2380** states are ever reached, and the 94 no-pure-saddle states carry **41%** of the discounted occupancy — a 2× over-representation on the path, 10× vs. the raw count (`soccer_nash/occupancy.py`, `docs/occupancy.md`). The carrier's optimal march to goal runs straight through the contested band; four of the eight most-visited states are mixed. The mixed structure is rare to *store* and common to *use*.

Freeze-then-iterate, and its staleness

RANDOM MOVE ORDER, HYBRID SOLVER — SAME FIXED POINT, $|DIFF| < 3E-9$

	ROUNDS / SWEEPS	MATRIX-GAME SOLVES	WALL CLOCK
value iteration	108	9 672	13.8 s
policy iteration	21	1 879	17.2 s

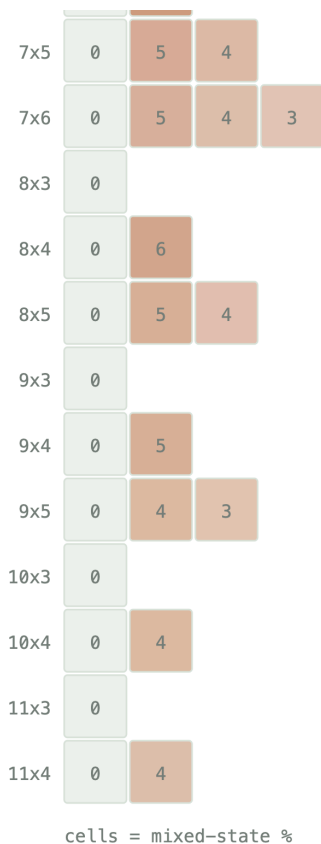
5× fewer LP solves — but the frozen strategies' distance to the current Nash stays *maximal* (a full pure-strategy flip) for **16 outer rounds**, then collapses to $< 1e-8$: concurrent stochastic games have no monotone policy improvement (`docs/discussion.md` §4). The thrashing eats the saving; on the deterministic game it is strictly worse. Value iteration with the per-sweep Nash cache wins here. A specific open question this ran into — which value quantity to back up in the frozen sweep, $p^T M q$ vs. the two pure bounds — is answered in `docs/numerics.md`

§3: the middle quantity converges in half the rounds (21 vs 41–42), because it is the unbiased estimate while the strategies are stale.

RQ3 – what characterizes the no-pure-saddle states

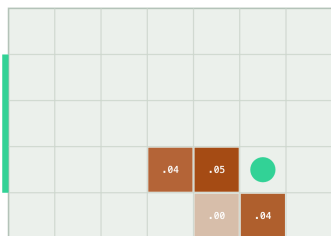
Across the tested family, a one-cell goal produced no no-pure-saddle stage games while every tested goal width ≥ 2 produced some. The phase diagram sweeps 127 configurations — boards with $3 \leq \text{width} \leq 11$, $3 \leq \text{height} \leq 9$, $\text{width} \cdot \text{height} \leq 45$, against goal widths 1 to height-2 (scripts/phase_diagram.py --analyze). Precisely: a multi-cell goal is *necessary* for a no-pure-saddle region in the studied family, and across the tested boards *sufficient* for at least one no-pure-saddle state — not a proof that every such board must have one, nor that every state is no-pure-saddle.

	goal-mouth width						
	1	2	3	4	5	6	7
3x3	0						
3x4	0	12					
3x5	0	10	9				
3x6	0	7	7	7			
3x7	0	7	6	6	6		
3x8	0	6	6	5	5	5	
3x9	0	6	5	5	4	4	4
4x3	0						
4x4	0	8					
4x5	0	8	6				
4x6	0	7	6	5			
4x7	0	7	5	5	4		
4x8	0	6	5	5	4	4	
4x9	0	6	5	4	4	4	3
5x3	0						
5x4	0	7					
5x5	0	6	4				
5x6	0	6	4	4			
5x7	0	5	4	4	3		
5x8	0	5	4	3	3	3	
5x9	0	5	4	3	3	3	3
6x3	0						
6x4	0	8					
6x5	0	6	4				
6x6	0	5	4	3			
6x7	0	4	4	4	3		
7x3	0						
7x4	0	7					



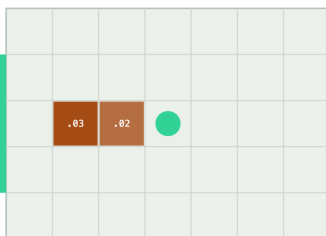
The gate (127 configs, $3 \leq W \leq 11$, $3 \leq H \leq 9$, $W \cdot H \leq 45$). Goal width 1 $\rightarrow$ 0 mixed stage games on all 40 tested one-cell configs. Goal width $\geq 2 \rightarrow$ at least one on all 87 tested wider-goal configs. The *fraction* (2.7–12%) is an OLS R^2 0.64 in board area – a dilution effect – goal width adding little with a negative coefficient.

defender at (5, 1)



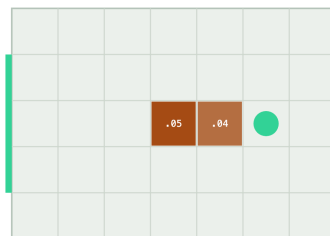
cells: minimax minus maximin of the carrier's stage game

defender at (3, 2)



cells: minimax minus maximin of the carrier's stage game

defender at (5, 2)



cells: minimax minus maximin of the carrier's stage game

Where the carrier has to guess. Defender pinned at the green disc; each cell shaded by minimax – maximin of the carrier's stage game (0 = pure saddle). Mixing is a local phenomenon – a handful of cells where the defender contests the forward lane.

Geometry predicts the label (scripts/geometry_model.py): a depth-4 decision tree on carrier-frame features separates mixed from the rest with **precision 0.96, recall 0.91**. The dominant feature is defender_can_intercept — the defender within one move of the carrier's forward cell ($P(\text{mixed}) = 0.44$ vs 0.002).

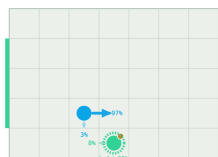
Beyond prediction, the 94 no-pure-saddle states canonicalize under the board mirror to 47 pairs and cluster into **8 geometric templates** (scripts/templates.py, docs/templates.md).

THE MIXED-STATE TEMPLATES – CARRIER-FRAME GEOMETRY, EQUILIBRIUM SUPPORT

STATES	GEOMETRY	SUPPORT	STRUCTURE
24	defender 1 ahead, carrier one row off the goal row	2×2	matching pennies
24	defender 1 ahead, same row, carrier on a goal row	2×2	matching pennies
16	defender 2 ahead, same row, carrier on a goal row	2×2	matching pennies
4	defender diagonally adjacent, carrier off the goal row	2×2	matching pennies
14	defender 1–2 ahead, same row, on a goal row	2×1	near-pure saddle
8	defender 2 ahead, same row, on a goal row	3×2	degenerate (row ties)
4	defender 2 ahead, carrier off the goal row	3×3	wider mix

The four 2×2-support templates are all verified matching pennies — the carrier's best reply flips between the two defender columns and vice versa — covering 68 of 94 states: the carrier choosing between two of {U, D, L, R} against the defender covering one of the same two (exact letters per template in docs/templates.md). All 94 share one value across every equilibrium; the forced-vs-degenerate split is below.

[1] 24 states

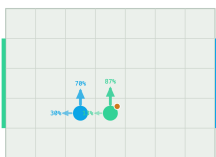
mixed equilibrium · $V = -0.150$

(2, 1, 3, 0, 1)

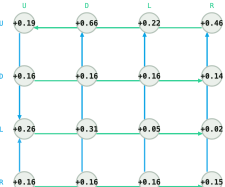


arrows cycle — no cell is a stable outcome, must mix

[2] 24 states

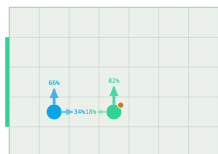
mixed equilibrium · $V = -0.199$

(2, 1, 3, 1, 1)

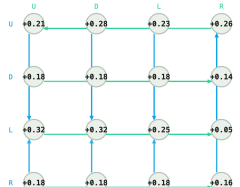


arrows cycle — no cell is a stable outcome, must mix

[3] 16 states

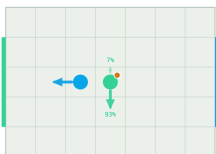
mixed equilibrium · $V = -0.225$

(1, 1, 3, 1, 1)



arrows cycle — no cell is a stable outcome, must mix

[4] 12 states

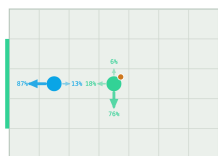
mixed equilibrium · $V = -0.211$

(2, 2, 3, 2, 1)

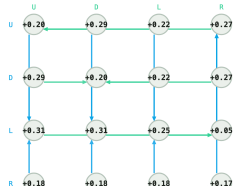


arrows cycle — no cell is a stable outcome, must mix

[5] 8 states

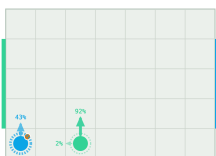
mixed equilibrium · $V = -0.220$

(1, 2, 3, 2, 1)

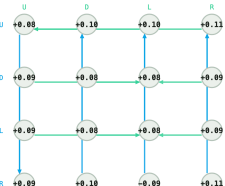


arrows cycle — no cell is a stable outcome, must mix

[6] 4 states

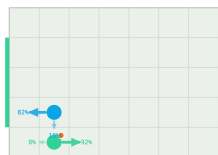
mixed equilibrium · $V = -0.005$

(0, 0, 2, 0, 0)

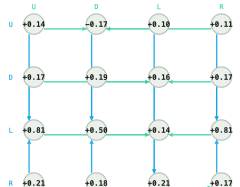


arrows cycle — no cell is a stable outcome, must mix

[7] 4 states

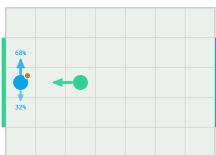
mixed equilibrium · $V = -0.206$

(1, 1, 1, 0, 1)

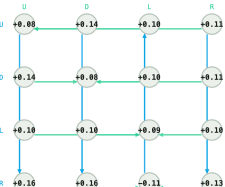


arrows cycle — no cell is a stable outcome, must mix

[8] 2 states

mixed equilibrium · $V = -0.103$

(0, 2, 2, 0, 0)



arrows cycle — no cell is a stable outcome, must mix

One policy fan per matching-pennies template. Each covers a class of no-pure-saddle states related by carrier-frame geometry; p^* , q^* are the equilibrium weights on the first support action. The carrier randomizes between two scoring lanes, the defender between covering them.

WHERE, AND HOW FORCED – THE TWO FACTS BEHIND THE FIGURE ABOVE

Where: all 94 no-pure-saddle states occur at Manhattan distance 1 or 2 between the players — 40 at distance 1, 54 at distance 2, **zero at distance ≥ 3** (scripts/distance_table.py, docs/showcase.md). Mixing is spatially localized around the opponent: all 94 no-pure-saddle states occur at player distance 1 or 2, while goal-mouth geometry determines whether competing scoring lanes exist in the first place.

How forced: a zero-weight action can tie the reported LP support's value exactly, so not every one of the 94 is a uniquely forced mix (scripts/degeneracy.py, docs/degeneracy.md):

THE 94 NO-PURE-SADDLE STATES, SPLIT BY HOW FORCED THE REPORTED MIX IS

CATEGORY	COUNT	SHARE
unique forced mix – no freedom left over	64	68.1%
degenerate equilibrium face – a zero-weight action ties the support	16	17.0%
pure reply tied inside the equilibrium set	14	14.9%

$$94 = 64 \text{ unique forced} + 16 \text{ degenerate faces} + 14 \text{ pure-tied}$$

The mechanism (machine-certified per config; interpretable, not a general theorem): a stage game requires mixing when a **stochastic ($\frac{1}{2}$ - $\frac{1}{2}$) resolution order** meets a **multi-cell goal** and a **defender that can contest the forward cell but not cover both lanes**. Then the carrier's "which lane" reply flips with the defender's cover and vice versa — *crossing best replies*, hence no pure saddle:

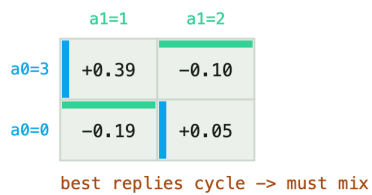
geometric condition $\Rightarrow$ crossing best replies $\Rightarrow$ no pure saddle

Drop any one clause — a one-cell goal, deterministic resolution, the coin-flip tie-break — and every tested stage game has a pure saddle. docs/mechanism.md gives the interpretable argument for the single-cell case. What is *not* settled is a board-size-free theorem for arbitrary W, H . *Supporting theory only* (not a project deliverable): for a single goal cell the defender's optimal strategy is closed-form and machine-verified, and every single-cell stage game is dominance-solvable, checked for all boards up to 11×5 .

A PER-STAGE-GAME CERTIFICATE

soccer_nash/certificate.py re-derives the pure/mixed label from the stage matrix with $O(A^2)$ arithmetic and no LP; scripts/verify_mechanism.py (make verify) runs it over the reachable state space. Single goal cell $\rightarrow$ a saddle-cell certificate at every state (0 mixed, cross-checked against onecell.certify). Wider goal $\rightarrow$ every mixed stage game gets a certificate an independent checker re-verifies to $\sim 1e-16$: the maximin $<$ minimax gap (the LP-free proof that no pure saddle exists), a closed best-reply cycle, the exact equilibrium ($\epsilon < 1e-15$), and a 2×2 **matching-pennies submatrix** — **present in all 94 of the 7×5 no-pure-saddle states and all 56 of the 5×4 ones**, strictly stronger than the 68-of-94 support count (the core is a property of the matrix; the LP's *reported* equilibrium can still be degenerate on 30 of the 94 without changing whether the core is there — docs/degeneracy.md). The switch is checked invariant across $\gamma \in \{0.5, 0.9, 0.995\}$, 4 vs 5 actions, win vs rate scoring, and exact undiscounted backward induction (16/16), and a $\pm 1e-6$ perturbation flips no genuine classification. docs/result.md is the precise statement and the prior-work positioning.

Mixed stage game: the 2×2 matching-pennies core



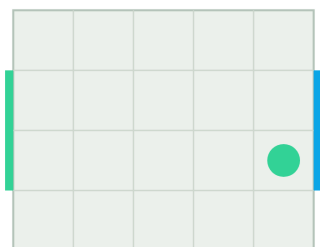
The 2×2 matching-pennies core of a mixed stage game. Blue bar = player 0's best row per column; green bar = player 1's best column per row. They never coincide — the best replies cycle, so mixing is forced. Every tested mixed stage game contains one.

The same fact, drawn as a node-and-arrow graph instead of a highlighted grid (one lit-up cell for a pure state, a closed loop of arrows for a mixed one) next to where the players actually are on the board, across twelve representative cases with the full 4×4 matrix printed for each — docs/positions.md / docs/positions.pdf, make positions. The deterministic A10 board gets its own edge-case document — every stage game there is provably pure, but ties, weak dominance, and the collision rule still hide real structure — a10_cases.pdf (make a10-cases-pdf); the coin-flip, tackle, territory-reward, and movement-slip rule variants each get the same treatment in coinflip_cases.pdf, tackle_cases.pdf, territory_cases.pdf, and slip_cases.pdf (make board-cases-pdfs), each case linking straight into the live explorer at that position.

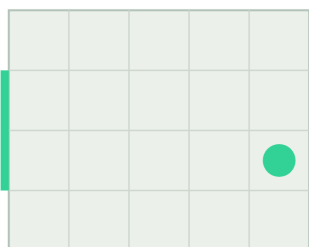
How far the goal-width switch generalizes (scripts/generalize.py, docs/generalize.md). It survives board scale (checked to 11×5 and 9×7), aspect ratio, and goal placement: non-contiguous goal cells like (1, 3) or (0, 2, 4) still produce no-pure-saddle stage games, so adjacency does not matter, only that the carrier has two cells the defender cannot cover at once. It does *not* survive a change of transition family. Adding action-independent movement noise (SoccerGame(slip=p): each player takes a random move with probability p) produces no-pure-saddle stage games even under deterministic resolution and even with a single goal cell — a 5×5 one-cell board

goes from 0 no-pure-saddle to 62 at `slip = 0.1`. The same is true of this project's own tackle rule — the defender commits to a challenge that wins the ball with probability `tackle_prob` or bounces off (docs/tackle.md). So the switch is a property of Littman's move-order rule, where the unresolved coin only bites where the carrier has two lanes; `slip` and `tackle` supply that coin everywhere, a fair coin flip on contested squares supplies it nowhere on the equilibrium path. Reading the 2×2 cores of the 7×5 no-pure-saddle states, the carrier's crossing pair is a vertical move in 90 of 94 — it is choosing which goal row to head for, and the defender is guessing it.

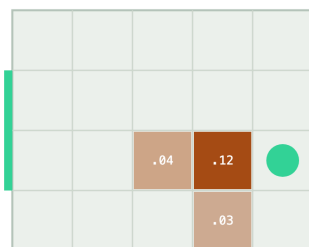
deterministic



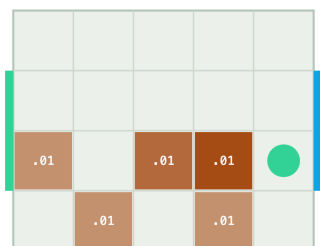
coinflip



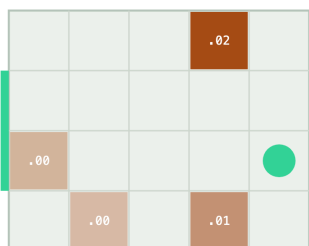
random



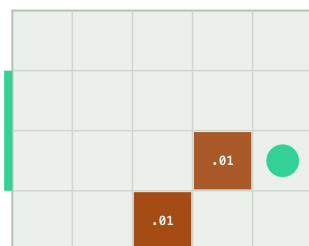
blend 0.7



slip 0.2



tackle 0.5



A fingerprint per collision rule. Same board, same pinned defender; shading is $\text{minimax} - \text{maximin}$ of the carrier's stage game. Each rule for resolving a contested cell leaves its own mark: the deterministic and coin-flip rules stay pure everywhere, Littman's move order and its blend light a thin band at the goal mouth, and the two rules that put an unavoidable coin on every square — `slip` and `tackle` — light a broad region around the defender.

EXISTENCE IS NOT THE SAME AS AMOUNT

Every one of the 94 no-pure-saddle stage games sits within 0.07 of a pure saddle (minimax – maximin, median 0.029), and the carrier's **mixing entropy** has median 0.55 bits — most are near-pure hedges ($\approx 87/13$), not real randomisation. Only **26 of 94** reach ≥ 0.9 bits (a near-even split); those are the states with genuine indifference. So the no-pure-saddle region is 94 states, but the *strongly*-mixed core is ~ 26 . Robust structure: 68 have a 2x2 matching-pennies support; the *value of mixing* ($V(\text{hybrid}) - V(\text{pure maximin})$) averages +0.13 there and compounds — at the centred kickoff a pure player secures only a draw where mixing is worth +0.15 (scripts/mixing.py).

The reward objective does not move the no-pure-saddle region

The default game rewards a win — first goal ends it, value $P(\text{win}) - P(\text{loss})$. A second objective, scoring="rate" (scripts/reward.py), keeps playing: a goal scores ± 1 and restarts with the ball to the conceding team, so the value is the expected discounted *goal difference* and $\gamma < 1$ is load-bearing. On the 7x5 random-order board:

SAME DYNAMICS, TWO REWARD OBJECTIVES – RANDOM MOVE ORDER, DISCOUNT 0.9

OBJECTIVE	NO-SADDLE STATES	VALUE RANGE	V(KICKOFF)
win – $P(\text{win}) - P(\text{loss})$	94	[–1.00, +1.00]	+0.150
rate – expected goal difference	94 – the same 94	[–0.88, +0.88]	+0.132

The no-pure-saddle set is *identical* — 0 states enter or leave. A mixed Nash equilibrium is an equilibrium of a single stage game $M(s)$; changing how the horizon is scored rescales $M(s)$ through $V(s')$ but does not cross the maximin = minimax boundary. The value *range* compresses ($\pm 1 \rightarrow \pm 0.88$): under rate a lost position is not a cliff — conceding restarts play with possession, a floor under every state. Value-surface figure and full detail in docs/reward.md.

Littman's fifth action

Littman's 1994 game adds a fifth action, stand; his Figure 2 — a carrier pinned against its own goal, forced to randomize between standing and moving — is the textbook "soccer needs mixing" example. Adding stand to the random-order game on Littman's 5x4 board (scripts/littman.py, full figure in docs/littman.md) does *not* move where mixing happens (48 of 56 no-pure-saddle states shared, 8 swap in and out at the margin) but changes *what* the mix is: the carrier now hedges with STAND in 48 of 56 states rather than a move action, and $V(\text{kickoff})$ improves $+0.147 \rightarrow +0.172$. The deterministic game keeps a pure saddle at every state with five actions too.

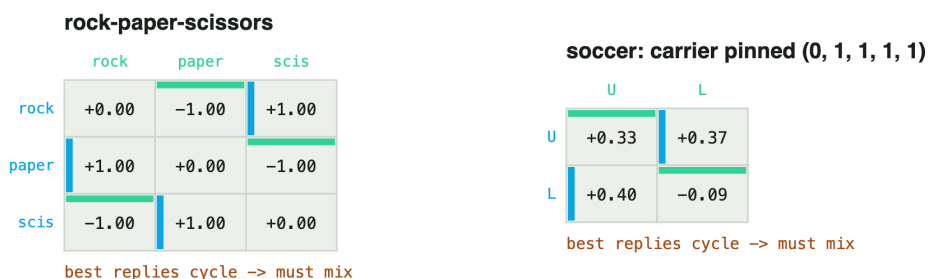
SO WHAT?

Knowing *where* the 94 no-pure-saddle states are and *why* they arise does not by itself say why a reader should care. "Why it matters" below answers that directly: a policy that plays a fixed, predictable action at those states can be learned and exploited by an opponent built specifically to punish it. That result is moved up into the reading guide near the top of this report for exactly this reason.

RQ4 – Numerical robustness

The soccer game contains rock-paper-scissors

The pure/mixed check is an $O(A^2)$ best-reply method: mark player 0's best row in each column and player 1's best column in each row; a cell with both is a pure saddle, and if none has both the best replies *cycle* and the game needs mixing — no LP required.



Same structure. Left: rock-paper-scissors. Right: the stage game at $(0, 1, 1, 1, 1)$ — carrier pinned against its own goal, defender adjacent (Littman's Figure 2 geometry). Blue bar = player 0's best row per column; green bar = player 1's best column per row. They cross both ways — matching pennies, no pure equilibrium.

The value is three numbers

`value_bracket` returns what the row player guarantees ($\min_k (pM)_k$), gets ($p^T Mq$), and can reach ($\max_i (Mq)_i$) — three quantities that are not guaranteed to be numerically identical. The true value is bracketed, so the width certifies the LP error.

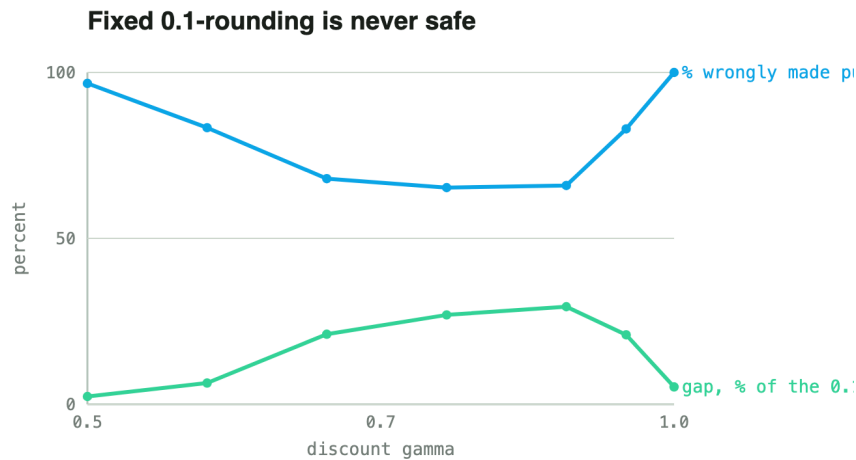
FINDING

With scipy's HiGHS the three agree to $1.1\text{e-}16$ per stage game, $8.7\text{e-}10$ accumulated — the definitional ambiguity is not a practical problem. If it were (an older vertex solver, or accumulated drift), the row player should take the *guaranteed* value $\min_k(pM)_k$: it is always a safe lower bound on the true minimax.

Discounting and tolerance

	DETERMINISTIC	RANDOM (7×5)
no-pure-saddle states, $\gamma = 0.5 \rightarrow 0.9 \rightarrow 0.995$	$0 \rightarrow 0 \rightarrow 0$	$122 \rightarrow 94 \rightarrow 102$
classification split, $\text{rel_tol } 1\text{e-}12 \rightarrow 1\text{e-}3$	stable	$604 / 1682 / 94$ (stable)
classification, $\text{rel_tol } 1\text{e-}2 / 1\text{e-}1$	—	$80 / 0$ mixed
fixed 0.1-rounding flips a saddle	0	62

Discounting mildly shifts which states mix; the classification is otherwise robust across nine orders of magnitude of tolerance, breaking only as the tolerance approaches the real minimax – maximin gaps. Fixed 0.1-rounding — proposed to force indifference — misclassifies 62 states, exactly the small-entry no-pure-saddle region.



There is no safe discount. The genuine minimax – maximin gaps (green) never reach the 0.1 rounding grid at any γ : heavy discounting shrinks the γ^k value differences, light discounting leaves entries barely apart. Fixed 0.1-rounding wrongly collapses 60–100% of the no-pure-saddle games (blue). The scale-aware `rel_tol` classifier is stable across the whole range.

Why it matters: minimax survives what every deterministic policy cannot

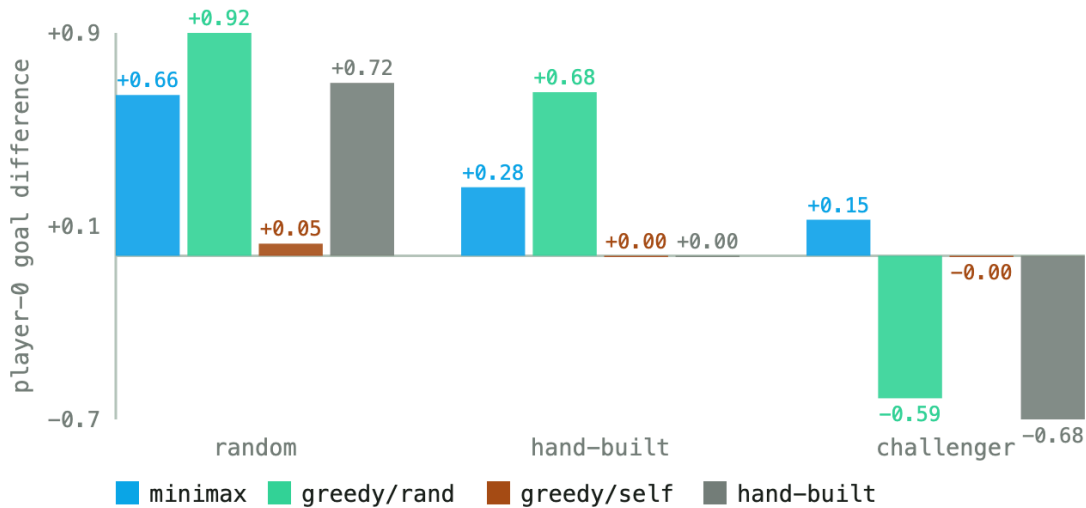
Littman's Table 3: minimax exploits *and* survives

Littman's central experiment trains four policies and scores each against a random opponent, a hand-built one ("simple rules for scoring and blocking", §6.2), and a **challenger** trained to beat it — his minimax policies held ~35–40% of games against their challenger, his Q-learning (deterministic) policies won 0%. Reproduced here with the *exact* solver on his 5×4 / 5-action / random-order / goal-reset board (`scripts/tournament.py`, `score = player-0 expected discounted goal difference`; the hand-built opponent beats a random one ~76% of games):

LITTMAN'S BOARD, $\Gamma = 0.9$ – PLAYER-0 GOAL DIFFERENCE FROM THE KICKOFF

PLAYER-0 POLICY	VS. RANDOM	VS. HAND-BUILT	VS. ITS CHALLENGER	ROBUSTNESS GAP
minimax (Nash)	+0.67	+0.28	+0.15	+0.52
greedy vs. random	+0.92	+0.68	−0.59	+1.51
greedy self-play	+0.05	0.00	−0.00	+0.05
hand-built	+0.72	0.00	−0.68	+1.39

Minimax holds up; greedy collapses to its challenger



Only minimax is in the top-left. greedy/rand is the strongest policy against weak opponents and among the weakest against a challenger – the rock-paper-scissors trap. minimax presses its advantage (+0.67) and still wins (+0.15) against the worst case, because the Nash policy has no exploitable row.

The mechanism is exactly Littman's "every deterministic offense has a perfect defense": the no-pure-saddle states are matching-pennies stage games; the greedy *and* the hand-built policy each commit to one row of each, and the challenger plays the column that beats it. The minimax policy randomizes with the weights that make the challenger indifferent. Full table (including the $P(\text{win}) - P(\text{loss})$ view) in docs/tournament.md.

THE MAIN PRACTICAL CONCLUSION OF THIS REPORT

Randomization matters not because it improves every matchup, but because it prevents a strong opponent from exploiting predictability. Greedy *beats* the minimax policy against a weak opponent (+0.92 vs. +0.67 above); it loses that advantage and more the instant something is built specifically to punish it (-0.59 vs. minimax's +0.15). The 94 no-pure-saddle states are exactly where a deterministic policy is predictable enough to hand a challenger that column.

docs/tournament_deepdive.md turns that mechanism into a measured, causal claim instead of an assertion: patching greedy's policy with the exact Nash mix at *only* the 7.4% of states that are genuinely mixed — leaving every other state as greedy already plays it — recovers 43% of the gap to minimax against a freshly built challenger, with one concrete exploited state shown matrix and all, and a gamma sweep showing that recovered share shrinks (and briefly reverses) as the horizon lengthens. This is the causal version of the conclusion above: the robustness gap traces specifically to the no-pure-saddle states, not to greedy play in general.

Other checks

- Duality gap $V0_{br}(s_0) + V1_{br}(s_0) < 1e-9$ for every move order — no opponent beats the game value.
- Nash vs. Nash reproduces the value: forced draws in the deterministic and coin-flip games; $+0.162 \pm 0.002$ empirical discounted return over 5 seeds of 3000 games (vs. $+0.164$ computed) in the random game.
- A best response to one assumed opponent is fragile: the Part 2 policy has exploitability 0.43 — worse than random — which is the A10 competition's warning about non-equilibrium submissions.
- A10 networks over 5 seeds (experiments/a10_competition_seeds.csv): the Part 2 imitation net plays optimally at 0.990 ± 0.000 of states and wins Q8 every time; competition Network Second is exact, but Network First sits at a 0.19 ± 0.02 exploitability from this kickoff that does not wash out with re-seeding.

Exploratory — function approximation and general-sum

These are groundwork for the eventual continuous-action work, not equal-weight contributions. The exact solver is the point of both: it is the ground truth they are measured against. Full tables and worked discussion for everything below: docs/neural.md.

Three starting points for the Q net (deterministic 7×5 game, 5 seeds): fitting a Q-net directly to the exact stage matrices, with no bootstrap at all, still only reaches **45% action agreement** — barely above the 43% a random init gets from 600 epochs of TD bootstrap. A policy net trained to match the equilibrium strategies directly reaches **99.9%** action agreement and is still the *most* exploitable of the four baselines tested (duality gap 0.84 vs 0.38–0.47) — the states it gets wrong are exactly the ones a best-responder attacks.

ACTION ACCURACY IS NOT EQUILIBRIUM ACCURACY

A network that names the textbook-optimal move nearly every time can still be trivially exploited, because the states where it is wrong are not random — they are exactly the ones a best-responder is looking for.

The harder game (random move order, genuine mixed equilibria, 5 seeds): the same lesson holds, sharper. The policy net reaches 95.8% action agreement and the *worst* exploitability of the four (1.20); fitting directly to the exact matrices is the closest numerical fit (60% agreement, lowest Frobenius error) and yet the second-most exploitable (1.01) — matching a continuous target well and matching a discrete equilibrium well are different objectives.

Capacity (width 32–512, depth 1–4, up to 115x the original parameter count): action agreement plateaus around 45–48% regardless of size, but exploitability keeps falling with capacity — 0.79 down to 0.05, a 15x reduction, with no sign of flattening. A bigger network keeps getting closer to the exact matrices in the sense that matters for robustness; it just never turns that into a better *argmax*.

General-sum (soccer_nash/markov_game.py, support_enum.py): pure-first extends beyond zero-sum by enumerating all stage equilibria and selecting one by largest sum of values, correctly handling Battle of the

Sexes; the soccer game itself is zero-sum, so this does not touch the main result. Future work, alongside the eventual continuous-action extension — see docs/neural.md's roadmap.

Limitations

PROOF STATUS, AT A GLANCE

Every numeric claim in this report falls into exactly one of three categories, so a swept parameter grid is never mistaken for a theorem (full wording: docs/assumptions.md §“Three claims, kept separate”).

STATUS	CLAIM
Proved	Deterministic A10 game: every stage game has a pure saddle – exact 100-step backward induction (238 000 state×step stage games) and stationary value iteration at every tested γ (Claim A).
Proved	The deterministic game has an explicit pure memoryless equilibrium – each player’s win-attractor strategy, verified to realize v^* at every state, goal widths 1 and 3 tested (Claim B).
Proved, single-cell case	Random move order, one goal cell per side: every stage game has a pure saddle. Defender’s optimal strategy closed-form and machine-verified; every board up to 11×5 dominance-solvable (Claim C’).
Empirical result	The goal-width switch – 1-cell goal → 0 no-pure-saddle stage games, ≥ 2 -cell → some – verified over 127 tested board/goal-width configurations. Not a theorem for arbitrary w, h .
Empirical result	The 94 → 47 mirror pairs → 8 templates collapse, the Manhattan-distance-1/2 geometry, and the 64/16/14 forced/degenerate split – measured on the canonical 7×5 board.
Open conjecture	A board-size-free proof that the carrier’s half of the single-cell defense is optimal for arbitrary w, h (the general-goal-mouth Claim C is not attempted at all).

- The A10 page redacts the ID-specific start position and goal rows; this repo uses the standard Littman geometry (configurable). Across the phase-diagram sweep every one-cell-goal board has 0 no-pure-saddle states and every wider-goal board has some; the exact 3.95% is the 7×5 / 3-cell-goal case.
 - Several collision sub-cases (carrier vs. stationary opponent, bump = steal, swap possession) are *interpreted* from the two stated A10 rules, not quoted. If course staff intended otherwise, Claim A must be re-measured. See docs/assumptions.md.
 - **Claim A/B** (a pure equilibrium of the deterministic game) is established constructively — the explicit pure memoryless attractor/safety profile. **Claim C** (the theoretical game necessarily has one, over all rules and settings) is *not*, for the general goal mouth; for the single goal cell it is partly proved (docs/proof.md).
 - random and coinflip are different game definitions, isolating the collision rule and the tie-break respectively — not the A10 game.
 - The goal-width result is **empirical over the tested grid**: a multi-cell goal is necessary for a no-pure-saddle region in the studied family and sufficient for at least one no-pure-saddle state on the tested boards — not a theorem.
-

What weight each experiment carries

Core. Pure vs. mixed by collision rule (RQ1); the pure-first hybrid, exact, LP eliminated on 96–100% (RQ2); the phase diagram — goal width, not board size (RQ3); geometry → templates → 2×2 matching pennies (RQ3); numerical robustness — the three value numbers, the rounding trap, the classifier (RQ4); Littman Table 3 reproduced and its causal follow-up — minimax exploits *and* survives a tailored challenger, greedy does one or the other, patching greedy with the exact mix at only the no-pure-saddle states recovers 43% of the gap (docs/tournament_deepdive.md) — the practical *why it matters*, not just a validation check.

Strong supporting evidence. Occupancy (4% of states, 41% of the path); the reward-objective comparison (region invariant, value range not); symmetry (value *and* policy equivariance); the blend interpolation; which no-pure-saddle states are a uniquely forced mix vs. degenerate (docs/degeneracy.md).

Future / exploratory. The neural baselines, deterministic and random-move-order games alike; the general-sum extension; freeze-then-iterate and the eval_value choice; the single-cell dominance / attractor machinery.

Questions and findings

One page, the questions this report set out to answer and the exact sections that answer them.

QUESTION	FINDING
Where does mixing occur?	94 of 2380 stage games (7×5 board, 3-cell goal, $\gamma = 0.9$) have no pure saddle, all at player distance 1-2 (RQ3)
Are these 94 distinct situations?	No – they canonicalize to 47 mirror pairs, collapsing to 8 geometric templates (RQ3)
What are the players actually doing?	Mostly choosing between two competing scoring/coverage lanes – 68 of 94 are 2×2 matching pennies (RQ3, docs/positions.md)
Are fractional LP outputs always a forced mix?	No – 64 uniquely forced, 16 degenerate equilibrium faces, 14 pure-tied (RQ3, docs/degeneracy.md)
Does mixing occur when players are far apart?	Never, at Manhattan distance ≥ 3 , in any tested configuration (RQ3)
Is randomness alone sufficient?	No – the coinflip variant stays pure everywhere; it is Littman's random move <i>order</i> that creates the coupling (RQ1)
Does goal width matter?	Yes, empirically: every tested 1-cell-goal board has 0 no-pure-saddle stage games, every tested ≥ 2 -cell board has some (RQ3)
Why does mixing matter in practice?	A predictable deterministic policy is exploitable by a challenger built specifically to beat it; minimax exploits weak play <i>and</i> survives its own challenger (“Why it matters,” C3)
Can neural approximation reproduce the equilibrium?	Not reliably – high action-naming accuracy does not imply low exploitability
Does a larger network fix that?	Capacity substantially lowers exploitability, but action agreement plateaus regardless of width or depth
What remains open?	A board-size-free proof for the single-cell case; the continuous-action extension

Related work

- **A. Markov games and minimax-Q.** Littman (1994) — the two-player zero-sum Markov game, minimax by LP, soccer *because* its optimal policy can be probabilistic. The baseline this project builds on.
- **B. Concurrent / reachability games.** de Alfaro–Henzinger–Kupferman on positional determinacy — the frame for the attractor result and the single-cell argument.
- **C. Stochastic games and stationary equilibria.** Shapley (1953); Filar & Vrieze; the orderfield-property literature — the frame for "one target cell vs. many".
- **D. Strategy improvement.** Hoffman–Karp, Condon for turn-based games; the *absence* of a monotone guarantee for concurrent games explains the freeze-then-iterate thrash.
- **E. Reward shaping.** Ng–Harada–Russell potential-based shaping — the frame for the PBRS-leaves-the-equilibrium-exact result.

Prior work: mixed strategies can be necessary in soccer (A). *This project adds:* an empirical map of where the no-pure-saddle region is, what switches it on, how much of the equilibrium path it covers, and a hybrid solver that pays LP cost only there (docs/result.md §6, full positioning against A–E).

soccer_nash — game.py · a10.py · matrix_games.py · support_enum.py · nash_q.py · markov_game.py · numerics.py · dominance.py · attractor.py · geometry.py · symmetry.py · tree.py · templates.py · onecell.py · certificate.py · reachability.py · occupancy.py · evaluate.py · best_response.py · exploit.py · mlp.py · nash_dqn.py · opponents.py · shaping.py · simulate.py · experiment.py · render.py · viz.py

scripts — experiments.py · analyze.py · numerics.py · equilibria.py · geometry_model.py · templates.py · phase_diagram.py · onecell_proof.py · verify_mechanism.py · generalize.py · tackle.py · positional.py · positions.py · benchmark.py · policy_iteration.py · selfplay.py · nash_dqn.py · nash_dqn_ablation.py · nash_dqn_random.py · degeneracy.py · distance_table.py · pure_vs_mixed_exploit.py · render.py · figures_png.py · gallery.py · story.py · showcase.py · littman.py · mixing.py · reward.py · blend.py · occupancy.py · tournament.py · tournament4.py · tournament_deepdive.py · a10_part1.py · a10_part2.py · a10_competition.py

docs — design.md · methods.md · discussion.md · numerics.md · assumptions.md · templates.md · proof.md · geometry.md · mechanism.md · result.md · generalize.md · tackle.md · littman.md · tournament.md · tournament_deepdive.md · positions.md · showcase.md · degeneracy.md · mixing.md · reward.md · blend.md · occupancy.md · findings.md · shaping.md · neural.md · selfplay.md · a10_part1.md · a10_part2.md · a10_competition.md · gallery.html

experiments — baseline.csv · undiscounted.csv · gamma_sweep.csv · tolerance_sweep.csv · board_sweep.csv · goal_mouth_sweep.csv · one_cell_goal.csv · phase_diagram.csv · generalize.csv · blend.csv · tackle.csv · reward.csv · mixing.csv · occupancy.csv · tournament.csv · tournament4.csv · tournament_deepdive.csv · degeneracy.csv · distance_table.csv · pure_vs_mixed_exploit.csv · nash_dqn_seeds.csv · nash_dqn_random_seeds.csv · nash_dqn_ablation.csv · a10_competition_seeds.csv

env: 7×5 grid, 2380 states, zero-sum ±1, $\gamma = 0.9$